Trends

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“The Embedded VISION SEGMENT Will Witness EXPLOSIVE GROWTH With Edge Processing!”

Indian OEM e-con Systems collaborates with chipset makers and sensor manufacturers to develop camera module solutions, targetting the US market. At India Electronics Week 2024, Co-founder Maharajan Veerabahu spoke to Yashasvini Razdan about the challenges and opportunities that India presents for the embedded vision segment and how e-con plans to navigate through this turf.

How difficult or easy is it to develop the firmware of a camera?

In a camera, you have a camera sensor, an image signal processor (ISP), and a lens. The firmware is primarily involved in the ISP, with some contribution in the sensor as well. The sensor frames need to be processed in the ISP, and the resulting output must pass through an interface to reach the host processor. The tuning of the ISP is an important part of firmware development, focusing on image signal processing. Understanding image processing and signal tuning is crucial for ISP development. On the other hand, if you are working on the protocol or interface side, an in-depth comprehension of the protocol and its implementation is crucial. We developed a USB video class (UVC) stack for USB 3.0 over two years, creating the stack from scratch and optimising it for speed and performance. In contrast, some ISP projects we undertook were more expedited, lasting only a couple of months. Each project’s development timeline varies based on its complexity and requirements.

Why don’t we see more and more Indian companies venturing in this space?

Usually, when system integrators encounter a problem, their initial approach often involves exploring existing options such as readily